Tide retrospective:
the $k = 2$ superPoly covariance bridge

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Abstract

The largest finding of the germbij fidelity review[1] was that the superPoly-language recovery headline *assumes* the Hessian equality that the note's Theorem 3.1 derives from the data: the merged Hessian step existed only in rate language (`hessian_recovery`, consuming covariance differences $o(q^2)$), with no bridge from the note's "same asymptotics" premise. This tide supplies that bridge. The new theorem[2] states: two localized losses whose second-moment families $t \mapsto \langle w_i w_j \rangle_t$ agree beyond all orders in the temperature have equal package matrices $H_1 = H_2$. The deliberation improved the candidate: no first-moment agreement is needed, because each package's first moments are $o(q)$ on their own, so the product terms in the covariance difference are $o(q^2)$ per package — only the second-moment difference consumes data. The proof is pure asymptotics plumbing over merged components and compiled on the first build.

1 Setting

The inverse half of germbij Theorem 3.1 was restated in the note's own data language by the expansion-bridge arc (PRs #93–#95), but its headline carries `hbase`: equality of derivatives of order 0–2, with both losses' domain packages sharing one matrix H. The fidelity review flagged this as the principal gap between the formalisation and the note's claim — the note *recovers* H from $t\ \mathrm{Cov}_t[w_i, w_j] \to (H^{-1})_{ij}$, and that step was formalised only against $o(q^2)$ *rate* hypotheses (stage H6's `hessian_recovery`), not against superpolynomial agreement of moment *families*. This tide is the first of three perimeter-closing steps (this bridge; the shift-normalization wrapper for the $j = 0$ constant; the cutoff-removal lemma), and is the minimal one: a single theorem, no new analysis, exactly the $k = 2$ analogue of the rate transport the expansion bridge already performs at $k > 2$.

2 The thing formalised

For localized losses L_1, L_2 carrying `LocalLaplaceDomain` packages with matrices H_1, H_2: if for every pair of coordinates i, j the difference of normalized localized second moments

$$t \longmapsto \langle w_i w_j \rangle_t^{L_1} - \langle w_i w_j \rangle_t^{L_2}$$

is smaller than every power of t^{-1} (the `SuperPoly` predicate, the note's $o(t^{-\infty})$), then $H_1 = H_2$. The Lean signature:

[1] `2026-08-10-01-29-review-germbij-new-content`.

[2] `hessian_recovery_of_superPoly_moments`, `Laplace/Multi/HessianBridge.lean`.

```
theorem hessian_recovery_of_superPoly_moments
    (A : LocalLaplaceDomain L₁ H₁) (B : LocalLaplaceDomain L₂ H₂)
    (hdata : ∀ i j : Fin d, Laplace.SuperPoly (fun t : ℝ ↦
      A.posteriorMomentT (fun w ↦ w i * w j) t -
        B.posteriorMomentT (fun w ↦ w i * w j) t)) :
    H₁ = H₂
```

Notably there is *no* first-moment hypothesis: the deliberation's strengthening (below). Combined with the merged `location_eq_of_superPoly_first_moments` and the $k > 2$ expansion bridge, all three moment layers of the note's Theorem 3.1 inverse now consume data in the same superPoly vocabulary; composing them into a single `hbase`-free headline is the next tide's job.

3 Proof strategy

Write $m^A(f)(q)$ for the normalized localized moment at scale q (temperature $t = q^{-2}$). The covariance form that `hessian_recovery` consumes is definitionally $m(w_i w_j) - m(w_i)\,m(w_j)$ — recorded as an `rfl` lemma (`covariance_eq_posteriorMoment`), since the merged file phrases it in raw `posteriorIntegral` quotients. The covariance difference then splits as

$$\left[m^A(w_iw_j) - m^B(w_iw_j)\right] \;-\; \left[m^A(w_i)m^A(w_j) - m^B(w_i)m^B(w_j)\right].$$

The first bracket is $o(q^2)$ by rate transport: the $m = 0$ case of the expansion bridge's `isLittleO_pow_of_superPoly` composed with the $t = q^{-2}$ substitution identity. For the second bracket no data is needed at all: the merged first-moment rate[3] ($m(w_i)/q \to 0$) converts to $m(w_i) = o(q)$ via `isLittleO_iff_tendsto'` (the $q \neq 0$ side condition holds eventually on $\mathcal{N}[>]0$), and `IsLittleO.mul` gives $m(w_i)m(w_j) = o(q \cdot q) = o(q^2)$ for each package separately. Assembling with `IsLittleO.sub` and a pointwise `ring` rearrangement feeds `hessian_recovery`, whose uniqueness-of-limits and positive-definite inversion argument does the rest.

4 Roadblocks and resolutions

None worth the name — the file compiled on the first `lake build`. The only design moment was the deliberation's strengthening:

> **GPT-5.6 Sol:** A can be strengthened and simplified: first-moment *agreement* is unnecessary. For each package independently, every first moment is $o(q)$, hence each product of first moments is $o(q^2)$. Therefore superpolynomial agreement of only the second moments already makes the covariance difference $o(q^2)$.

This is strictly less hypothesis for the same conclusion and was adopted as the agreed candidate A′. It also has a pleasant reading: at the Hessian layer, the first moments are pure gauge — their information content ($o(q)$ smallness) is a consequence of localization, not of the data.

5 What was Mathlib, what was new

Mathlib lookup: `isLittleO_iff_tendsto'`, `IsLittleO.mul/sub/congr'`, `sq`. Seabed reuse: `hessian_recovery` (H6), `tendsto_normalized_first_moment` (H5), `isLittleO_pow_of_superPoly`

[3] `tendsto_normalized_first_moment`, `HessianMoments.lean`.

and `posteriorMomentT_inv_sq` (expansion bridge). New: only the four declarations of the file — the `rfl` vocabulary bridge, the two little-o packagings, and the headline composition. This is the intended shape for a perimeter tide: zero new analysis, one new composed statement.

6 Lessons

The $m = 0$ case of `isLittleO_pow_of_superPoly` (transport with no power division) is the workhorse for restating *any* rate-language recovery theorem in superPoly data language; tides 2 and 3 of this perimeter should reach for it first. And the review pipeline worked as designed: finding F1's "no $k = 2$ analogue of the expansion bridge exists" was a precise, one-tide-sized gap statement.

7 Follow-ups

- Tide 2 (next): the shift-normalization wrapper discharging `hbase` at $j = 0$, plus the recomposition: an `hbase`-free headline concluding positive-order jet equality from superPoly data alone, using this tide's $H_1 = H_2$ to instantiate the shared-H machinery, a `HigherLaplaceDomain`-level $T_2 = \text{qform}/2$ tie for the $j = 2$ tensor form, and the $j = 1$ vanishing from the quadratic Peano field.
- Tide 3: the cutoff-removal lemma (C_c^∞ data implies monomial data), completing the note-literal premise.